

OGMP: Oracle Guided Multi-mode Policies for Agile and Versatile Robot Control

Lokesh Krishna, Nikhil Sobanbabu, and Quan Nguyen

Abstract—The efficacy of reinforcement learning for robot control relies on the tailored integration of task-specific priors and heuristics for effective exploration, which challenges their straightforward application to complex tasks and necessitates a unified approach. In this work, we define a general class for priors called oracles that generate state references when queried in a closed-loop manner during training. By bounding the permissible state around the oracle’s ansatz, we propose a task-agnostic oracle-guided policy optimization. To enhance modularity, we introduce task-vital modes, showing that a policy mastering a compact set of modes and transitions can handle infinite-horizon tasks. For instance, to perform parkour on an infinitely long track, the policy must learn to jump, leap, pace, and transition between these modes effectively. We validate this approach in challenging bipedal control tasks: parkour and diving—using a 16-DoF dynamic bipedal robot, HECTOR. Our method results in a single policy per task, solving parkour across diverse tracks and omnidirectional diving from varied heights up to $2m$ in simulation, showcasing versatile agility. We demonstrate successful sim-to-real transfer of parkour, including leaping over gaps up to 105% of the leg length, jumping over blocks up to 20% of the robot’s nominal height, and pacing at speeds of up to 0.6 m/s, along with effective transitions between these modes in the real robot.

I. INTRODUCTION

Deep reinforcement learning (RL) has shown remarkable success in synthesizing control policies for hybrid and underactuated legged robots [1], particularly in enabling inherently stable quadrupedal robots to achieve extreme parkour [2]–[4], agile [5], and robust [6] locomotion. Following a common philosophy: 1) define an exhaustive observation space, 2) engineer task-specific rewards and/or curriculum, 3) perform policy distillation, and 4) extensively randomize, these methods rely on task-specific tricks in each step, lacking a systematic approach for robot control. Specifically, since Deep RL methods are quasi-solvers for unconstrained optimization, they are prone to anomalous, case-specific local optima. Hence, practitioners often resort to task-specific reward shaping [2], [5], [6] and heuristics [3] for a structured exploration and to meet the intended performance. Furthermore, the established approach for robot control is privileged training and policy distillation: training teacher policies [3], [4] with privileged information, solving a pseudo-MDP with RL, and distilling it into a single policy for the true POMDP. In contrast, we aim to find an optimal policy by structured exploration in the true POMDP guided by priors.

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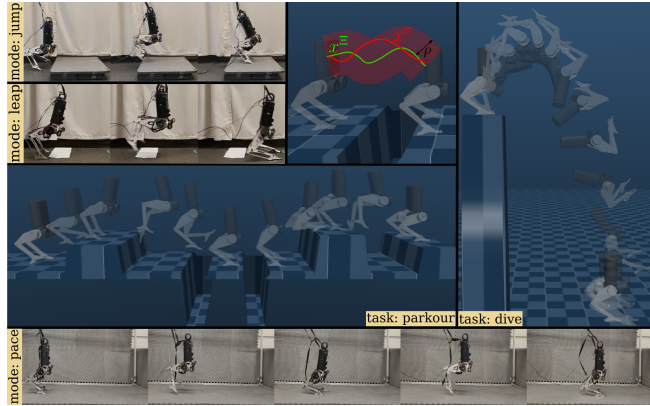


Fig. 1: Overview of OGMP: Oracle guided policy optimization and the applied tasks visualized. Trained OGMPs π_{parkour} performing agile parkour in the simulation and the real-robot. π_{dive} performing a frontflip dive from a 2m high platform in simulation. Accompanying video results at : [link](#)

Guided Policy Optimization (GPO) aims to improve sample efficiency and mitigate poor local optima. GPO methods are either *Control-guided* or *Reference-guided*. *Control-guided* approaches require control trajectories: [7], [8] employ prior controllers and policy-trajectory constraints, while [9], [10] alternate trajectory optimization and minimize RL variance. [11]–[13] demonstrate quadrupedal locomotion but relying on pre-existing model-based controllers, limiting complex tasks (e.g., parkour). *Reference-guided* methods like [14]–[18] use morphologically similar state-reference trajectories to guide RL to learn the corresponding optimal actions for character control. However, with pre-generated open-loop reference, policy exploration is confined to the demonstration’s scenario, thus hindering the emergent behaviors (like recovery) seen in from-scratch RL methods [1]–[3], which explore full-order dynamics and challenging randomization in simulation, crucial for real robot control. Therefore, we propose a reference-guided policy optimization using closed-loop state-reference generators (oracles) that can be queried dynamically to produce references from any state, with a novel hyperparameter to address local optima in complex tasks.

Alternatively, imitation learning (IL) proves to be a reliable task-agnostic strategy for robot manipulation with dynamically consistent demonstrations through proficient human teleoperation, which solves the intended task [19]. In contrast, we have dynamically inconsistent demonstrations for locomotion that partially solve a task, challenging the direct application of IL. For instance, to parkour with a bipedal robot, we may have demonstrations for runs, leaps, and

OGMP: 面向敏捷与多功能机器人控制的 Oracle 引导多模式策略

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摘要 — 强化学习在机器人控制中的有效性依赖于任务特定先验知识和启发式方法的定制化整合, 以实现高效探索, 这给其直接应用于复杂任务带来了挑战, 并需要一种统一的方法。在这项工作中, 我们定义了一类通用的先验知识, 称为 Oracle, 它在训练过程中以闭环方式被查询时生成状态参考。通过将允许的状态限制在 Oracle 假设的范围内, 我们提出了一种任务无关的 Oracle 引导策略优化方法。为了增强模块化, 我们引入了任务关键模式, 表明掌握一组紧凑模式及其转换的策略可以处理无限时域任务。例如, 要在无限长的跑道上完成跑酷, 策略必须学会跳跃、跨越、踱步以及在这些模式之间有效转换。我们使用 16 自由度的动态双足机器人 HECTOR, 在具有挑战性的双足控制任务 (跑酷和跳水) 中验证了该方法。我们的方法为每个任务生成单一策略, 在仿真中解决了跨不同轨道的跑酷以及从高达 $2m$ 的不同高度进行的全向跳水问题, 展示了多功能的敏捷性。我们展示了跑酷从仿真到真实世界的成功迁移, 包括跨越高达腿长 105% 的间隙、跳过高达机器人标称高度 20% 的障碍物、以高达 0.6 米 / 秒的速度踱步, 以及在真实机器人上这些模式之间的有效转换。

I. 引言

深度强化学习在合成混合驱动和欠驱动腿式机器人的控制策略方面取得了显著成功 [1], , 特别是在使天生稳定的四足机器人实现极限跑酷 [2]–[4], 敏捷 [5], 且鲁棒 [6] 的运动方面。遵循一种常见理念: 1) 定义详尽的观测空间, 2) 设计任务特定的奖励和 / 或课程, 3) 进行策略蒸馏, 以及 4) 广泛随机化, 这些方法在每个步骤中都依赖于任务特定的技巧, 缺乏系统化的机器人控制方法。具体而言, 由于深度强化学习方法是无约束优化的准求解器, 它们容易陷入异常且针对特定情况的局部最优。因此, 实践者常常借助任务特定的奖励塑造 [2], [5], [6] 和启发式方法 [3] 来实现结构化探索并达到预期性能。此外, 机器人控制的既定方法是特权训练和策略蒸馏: 利用特权信息训练教师策略 [3][4], 通过强化学习求解伪马尔可夫决策过程, 并将其蒸馏为针对真实部分可观测马尔可夫决策过程的单一策略。相比之下, 我们的目标是通过在真实部分可观测马尔可夫决策过程中进行由先验知识引导的结构化探索来寻找最优策略。

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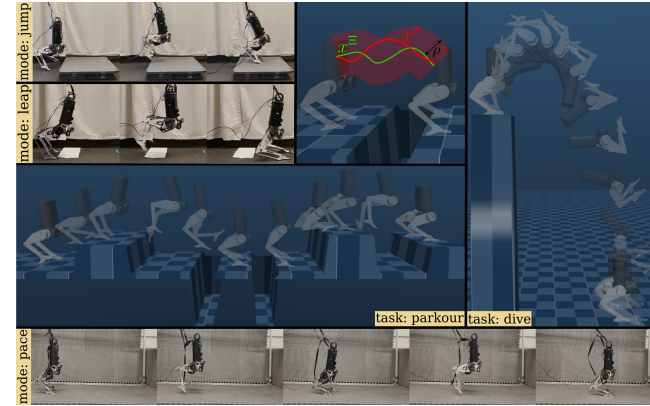


图 1: OGMP 概述: Oracle 引导的策略优化及所应用任务的可视化。训练后的 OGMP π_{parkour} 在仿真和真实机器人上执行敏捷跑酷。 π_{dive} 在仿真中从 2 米高平台执行前空翻跳水。附带的视频结果位于: [link](#)

引导策略优化 (GPO) 旨在提高样本效率并缓解不良局部最优问题。GPO 方法要么是控制引导型或参考引导型。控制引导型方法需要控制轨迹: [7], [8] 采用先验控制器和策略 - 轨迹约束, 同时 [9], [10] 交替进行轨迹优化并最小化强化学习方差。[11]–[13] 展示了四足运动, 但依赖于预先存在的基于模型的控制, 限制了复杂任务 (如跑酷)。参考引导型方法如 [14]–[18] 使用形态相似的状态参考轨迹来引导强化学习学习相应的最优动作, 用于角色控制。然而, 使用预先生成的开环参考时, 策略探索被限制在演示场景内, 从而阻碍了从零开始的强化学习方法 [1]–[3], 中出现的涌现行为 (如恢复), 这些方法探索全阶动力学和仿真中具有挑战性的随机化, 这对真实机器人控制至关重要。因此, 我们提出了一种使用闭环状态参考生成器 (oracle) 的参考引导策略优化, 该生成器可从任何状态动态查询生成参考, 并引入一种新颖的超参数来解决复杂任务中的局部最优问题。

或者, 模仿学习 (IL) 被证明是一种可靠的任务无关策略, 用于通过熟练的人类遥操作提供动态一致的演示来实现机器人操作, 从而解决预期任务 [19]。相比之下, 我们用于运动的演示是动态不一致的, 只能部分解决任务, 这对直接应用模仿学习构成了挑战。例如, 要让双足机器人完成跑酷, 我们可能有跑步、跳跃和

jumps from motion capture on humans, which suffer from morphological dissimilarity due to source-target mismatches [20]. Moreover, naive imitation of partial demonstrations (run, leap, etc.) does not guarantee to solve the overarching task (parkour), requiring high-level RL-trained policy for transitions and emergent behaviors [20]–[22]. Besides demonstrations, robot tasks have rich priors like heuristics [23], task/motion planners [24], and model-based controllers [25], which can guide learning, leading to regularized behaviors [13]. While the idea of imitating such priors has been studied, we instead propose building a “trust region” in the state space around the prior’s solution. Thus, the more we “trust” a prior, the tighter the trust region could be and vice versa. Formally defining a general class for priors: oracle, an oracle-guided policy optimization can be performed by bounding the policy’s permissible state space within the local neighborhood of an oracle’s ansatz. Empirically, we observe that the right choice of this bound helps escape erroneous local optima providing an optimal balance between emergent and regularized behaviors ideal for robot control, making it an effective hyperparameter in practice.

On the other hand, solving complex tasks requires behavioral multi-modality. Classical multi-mode control [26], [27] involves switching among a finite set of pre-designed controllers to address high-level tasks, creating a pseudo-hybrid system. Learning methods leave multi-modality in control to emerge implicitly [3], [5], lacking a methodical synthesis. [22] propose encoding a dataset of demonstrations to a latent space and latent conditioning to train multi-skilled policies. However, the notion of skill is poorly defined. For instance, solving a task requires not only mastering discrete modes (e.g., walking, jumping) but also continuous parameter variation of the same (e.g., speed, height) and inter-skill transitions. [2] trains multiple low-level controllers and a high-level mode-switching policy for quadrupedal parkour, requiring diverse reward shaping and training routines. [28] shows that a “fixed” set of uni-mode controllers limits complex transition maneuvers, introducing a single multi-mode policy for mastering a set of modes and transitions to handle new tasks zero-shot. In line with this approach, we aim to achieve a single policy that learns a finite set of modes with “infinite” parameter variations and transitions through reference-guided policy optimization. Unlike switching fixed controllers, we hypothesize that reference-guided exploration can better accommodate emergent control modes. To this end, we first show task-vital multi-modality as a way to decompose tasks into their principal modes and transcribe them into our proposed OGMP framework. Thus, the major contributions of our paper are twofold:

- Oracle Guided Multi-mode Policies: A theoretical framework for task-centered control synthesis leveraging oracle-guided optimization to effectively search through bounded exploration and task-vital multi-modality for versatile control.
- Experimental validation on agile bipedal control tasks requiring versatility, such as parkour and diving. A

single policy per task demonstrates the ability to perform diverse variants of the task-vital modes realized in simulation and the real-robot on the 16-Dof bipedal robot Hector.

The remaining paper is structured as follows: Sec. II presents the theoretical development of our framework, Sec. III discusses applying the proposed framework to bipedal control tasks, followed by Sec. IV presenting the experimental results, analysis and ablation studies.

II. ORACLE GUIDED MULTI-MODE POLICIES

This section presents our theoretical framework with two synergetic ideas: oracle-guided policy optimization and task-vital multi-modality. Specifically, we aim to prune undesirable local optima by bounding exploration to the local neighborhood of an oracle and by designing the learning of multiple behavior modes and transitions to effectively solve tasks.

1) *Oracle Guided Policy Optimization(OGPO)*: Let $\mathcal{T}$ be an infinite horizon task with a task parameter set, $\psi_{\mathcal{T}} \in \Psi_{\mathcal{T}}$. Sufficiently solving $\mathcal{T}$ requires maximizing a task objective, $J_{\mathcal{T}}$ over the task parameter distribution, $p(\psi_{\mathcal{T}})$. Given the corresponding state space (or task space) of interest, $x \in \mathcal{X}$ let x_t , $x[a, b]$ denote a state at time t and a state trajectory from time $t \in [a, b]$ respectively. We define Ξ to be a receding horizon oracle that provides a finite horizon state trajectory ($x^{\Xi}[t, t + \Delta t]$) from any given state (x_t) until Δt into the future for any task-variant ($\psi_{\mathcal{T}}$), such that $x^{\Xi}[t, t + \Delta t]$ is always within an ϵ -neighborhood of an optimal state trajectory. Formally,

$$x^{\Xi}[t, t + \Delta t] = \Xi(x_t, \psi_{\mathcal{T}}) \quad (1a)$$

$$\text{s.t. } \exists x^{\Xi}[t, t + \Delta t] \quad \forall (x_t, \psi_{\mathcal{T}}) \quad (1b)$$

$$\|x_t^{\Xi} - x_t^*\|_W < \epsilon \quad \forall t \in [0, \infty) \quad (1c)$$

where ϵ is the maximum deviation bound, a constant for a given pair $(\mathcal{T}, \Xi)$, and W is a diagonal weight matrix¹. We aim to obtain the optimal policy π^* from a policy class Π guided by Ξ that sufficiently solves $\mathcal{T}$. Since Ξ provides a reference in the state space, we propose constraining the permissible states for π to be within a ρ -neighbourhood of the oracle’s guidance. Formally,

$$\pi^* := \arg \max_{\pi \in \Pi} J_{\mathcal{T}} \quad (2a)$$

$$\text{s.t. } \|x_t^{\pi} - x_t^{\Xi}\|_W < \rho \quad \forall t \in [0, \infty) \quad (2b)$$

Where x^{π} are the states visited while rolling out policy π , ρ is the permissible state-bound for oracle-guided exploration. With the above setup, one can observe that the bounded set of permissible states visualized in Fig. 2i is given by

$$x_t^{\pi} \in \{x \mid \|x_t^{\Xi} - x_t^*\| - \rho \leq \|x - x_t^*\| \leq \|x_t^{\Xi} - x_t^*\| + \rho\} \quad (3)$$

Thus, ρ should be chosen so that x_t^* is reachable by x_t^{π} , which requires the lower bound in Eq. 3, to be non-positive. Therefore as shown in Fig 2ii, for x_t^* to be within

¹Note that $x^*[0, \infty]$, ϵ are unknown and are only provided for constructing the conceptual argument

从人类运动捕捉中跳跃存在源 - 目标不匹配导致的形态学差异问题 [20]。此外，对部分演示（如奔跑、跳跃等）的简单模仿并不能保证解决总体任务（如跑酷），这需要经过强化学习训练的高级策略来实现过渡和涌现行为 [20]–[22]。除了演示之外，机器人任务还拥有丰富的先验知识，如启发式方法 [23]、任务 / 运动规划器 [24]，和基于模型的控制器 [25]，这些都可以指导学习，从而产生规范化的行为 [13]。虽然模仿此类先验的想法已有研究，但我们提出在状态空间中围绕先验解构建一个“信任区域”。因此，我们对先验的“信任”程度越高，信任区域就可以越紧凑，反之亦然。通过形式化定义一类通用的先验：预言机，可以通过将策略的允许状态空间限制在预言机假设解的局部邻域内，来执行预言机引导的策略优化。实验表明，正确选择这一边界有助于逃离错误的局部最优解，在涌现行为与规范化行为之间实现理想平衡，非常适合机器人控制，使其在实践中成为一个有效的超参数。

另一方面，解决复杂任务需要行为的多模态性。经典的多模态控制 [26][27] 涉及在一组有限的预设计控制器之间切换以处理高级任务，从而创建伪混合系统。学习方法让控制中的多模态性隐式涌现 [3], [5]，缺乏系统性的综合方法。[22] 提出将演示数据集编码到潜在空间，并通过潜在条件训练多技能策略。然而，技能的概念定义不清。例如，解决一个任务不仅需要掌握离散模式（如行走、跳跃），还需要同一模式的连续参数变化（如速度、高度）以及技能间的过渡。[2] 为四足机器人跑酷训练了多个低级控制器和一个高级模式切换策略，这需要多样化的奖励塑造和训练流程。[28] 表明，“固定”的单模态控制器集合限制了复杂的过渡机动，因此引入了一个单一的多模态策略来掌握一组模式及其过渡，以零样本方式处理新任务。与此方法一致，我们旨在通过参考引导的策略优化，实现一个单一策略，该策略学习有限数量的模式，同时具备“无限”的参数变化和过渡能力。与切换固定控制器不同，我们假设参考引导的探索能更好地适应涌现的控制模式。为此，我们首先将任务关键的多模态性作为将任务分解为其主要模式的方法，并将其转录到我们提出的 OGMP 框架中。因此，本文的主要贡献有两点：

- Oracle 引导的多模式策略：一种面向任务中心控制综合的理论框架，利用 Oracle 引导的优化，通过有界探索和任务关键的多模态性有效搜索，实现多功能控制。
- 在需要多功能性的敏捷双足控制任务（如跑酷和跳水）上的实验验证。

每个任务使用单一策略，展示了执行任务关键模式多种变体的能力，这些模式在仿真和 16 自由度双足机器人 Hector 的真实机器人上得以实现。

本文其余部分结构如下：第 II 节介绍我们框架的理论发展，第 III 节讨论将所提框架应用于双足控制任务，随后第 IV 节展示实验结果、分析和消融研究。

II. ORACLE 引导的多模式策略

本节介绍我们的理论框架，包含两个协同思想：Oracle 引导的策略优化和任务关键的多模态性。具体而言，我们旨在通过将探索限制在 Oracle 的局部邻域内，并设计多种行为模式及其转换的学习，来有效剪除不期望的局部最优解，从而有效解决任务。

1) Oracle 引导的策略优化（OGPO）：设 $\mathcal{T}$ 为一个具有任务参数集 $\psi_{\mathcal{T}} \in \Psi_{\mathcal{T}}$ 的无限时域任务。充分求解 $\mathcal{T}$ 需要在任务参数分布 $p(\psi_{\mathcal{T}})$ 上最大化任务目标 $J_{\mathcal{T}}$ 。给定相关的状态空间（或任务空间）， $x \in \mathcal{X}$ 令 x_t , $x[a, b]$ 分别表示时刻 t 的状态和从时刻 $t \in [a, b]$ 开始的状态轨迹。我们将 Ξ 定义为一个滚动时域 Oracle，它从任意给定状态（ x_t ）出发，针对任意任务变体（ $\psi_{\mathcal{T}}$ ），提供未来直到 Δt 的有限时域状态轨迹（ $x^{\Xi}[t, t + \Delta t]$ ），使得 $x^{\Xi}[t, t + \Delta t]$ 始终处于最优状态轨迹的 ϵ -邻域内。形式上，

$$x^{\Xi}[t, t + \Delta t] = \Xi(x_t, \psi_{\mathcal{T}}) \quad (1a)$$

$$\text{s.t. } \exists x^{\Xi}[t, t + \Delta t] \quad \forall (x_t, \psi_{\mathcal{T}}) \quad (1b)$$

$$\|x_t^{\Xi} - x_t^*\|_W < \epsilon \quad \forall t \in [0, \infty) \quad (1c)$$

其中 ϵ 是最大偏差界，对于给定对 $(\mathcal{T}, \Xi)$ 是一个常数，而 W 是一个对角权重矩阵¹。我们的目标是在 Ξ 的引导下，从策略类 Π 中获得能够充分求解 $\mathcal{T}$ 的最优策略 π^* 。由于 Ξ 在状态空间中提供了参考，我们建议将 π 的允许状态约束在 Oracle 引导的 ρ -邻域内。形式上，

$$\pi^* := \arg \max_{\pi \in \Pi} J_{\mathcal{T}} \quad (2a)$$

$$\text{s.t. } \|x_t^{\pi} - x_t^{\Xi}\|_W < \rho \quad \forall t \in [0, \infty) \quad (2b)$$

其中 x^{π} 是执行策略 π 时访问的状态， ρ 是 Oracle 引导探索的允许状态边界。通过上述设置，可以观察到图 2i 中可视化的有界允许状态集由下式给出

$$x_t^{\pi} \in \{x \mid \|x_t^{\Xi} - x_t^*\| - \rho \leq \|x - x_t^*\| \leq \|x_t^{\Xi} - x_t^*\| + \rho\} \quad (3)$$

因此，应选择 ρ 使得 x_t^* 对于 x_t^{π} 是可达的，这要求式 3 中的下界为非正数。因此如图 2ii 所示，为使 x_t^* 处于

¹注意， $x^*[0, \infty]$, ϵ 是未知的，仅用于构建概念性论证

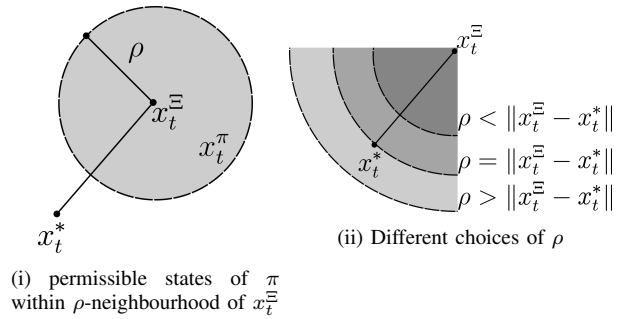


Fig. 2: Oracle bounded exploration

the permissible states of π , ρ must satisfy

$$(x_t^* \text{ is reachable by } x_t^\pi) \implies \rho \geq \|x_t^\pi - x_t^*\| \quad (4)$$

By definition (Eq. 1.c), since $\|x_t^\pi - x_t^*\| < \epsilon$ a sufficient choice of ρ to satisfy Eq 4 for all time, is

$$\rho \geq \epsilon \quad (5)$$

ϵ being maximum deviation of x^π , an oracle with low ϵ will generate references closer to x^* thus, the exploration can be bounded to tight-neighborhood to filter out most local optima in the objective landscape. In contrast, for a “poor” oracle with high ϵ , there should be sufficient search space for π to explore around x^π and converge to x^* . From Eq 5, as $\epsilon \rightarrow \infty$, the optimization is unguided, thus needing $\rho \rightarrow \infty$ (a standard RL setting). Conversely, as $\epsilon \rightarrow 0$, an arbitrarily small ρ satisfying Eq 5 could be chosen to avoid local optima by localizing the search while still being able to converge x_t^* . In practice, as ϵ is unknown, we perform a grid search over ρ for any given $(\mathcal{T}, \Xi)$.

2) *Task Vital Multi-modality*: A policy learning to solve a task, $\mathcal{T}$, can be seen as mastering a “bundle” of spacetime trajectories in the task space, corresponding to $\psi_{\mathcal{T}} \in \Psi_{\mathcal{T}}$. Simple tasks allow straightforward oracle construction satisfying Eq 1. However, complex and infinite-horizon tasks make $\Psi_{\mathcal{T}}$ intractable. For instance, an oracle for indefinite parkour requires knowledge of an infinite track apriori, which is impractical. To address this, we define modes as finite spacetime segments that preserve some spatial and/or temporal invariances. Like in parkour, modes like jump and leap remain consistent regardless of location or time. Therefore, we define a finite set of modes, $\mathbb{M}$, having a temporal length of Δt , vital for $\mathcal{T}$. Each $m \in \mathbb{M}$ (like jump) can have continuous parameters Ψ_m (like jump height). Then the mode parameter set $\Psi_{\mathbb{M}} := \bigcup_{i=1}^{|\mathbb{M}|} \Psi_{m_i}$ is related to the task parameter set as $\Psi_{\mathcal{T}} := \Psi_{\mathbb{M}}^N$, where N is the number of horizons. Thus, by mastering modes in $\mathbb{M}$ (jump, leap, and pace) and transitions over varying $\Psi_{\mathbb{M}}$ (speeds, distances, and heights), task $\mathcal{T}$ (indefinite parkour, $N \rightarrow \infty$) can be solved as visualized in Fig 3.a.

III. DESIGN METHODOLOGY

This section presents the design methodology using OGMP for the bipedal control tasks: parkour and dive, as shown in Fig 3. For a given task, we first define the task-vital modes—such as jump, leap, and pace for parkour—and design a reference generation scheme for each

$\mathcal{T} : J_{\mathcal{T}}$	$(m, \Psi_m) \in (\mathbb{M}, \Psi_{\mathbb{M}})$	
	Mode	Parameters
dive: 360° flip and land	settle flip	$\{\}$ $\{(r, h)\}$
parkour: traverse the track indefinitely	pace jump leap	$\{v\}$ $\{(w, h)\}$ $\{(w, d)\}$

TABLE I: Task description and corresponding task vital multi-modality. Parameters visualized in Fig. 4

(Fig. 3.a). Spanning the mode parameter set, we employ the oracle to generate a custom dataset of diverse behaviors and train a mode encoder to construct a compact latent space for command conditioning (Fig 3.c). Finally, we train a multi-mode policy guided by the oracle (Fig 3.d). During policy optimization, the oracle is periodically queried to generate references online, bounding the policy’s search space to a reference’s local neighborhood for effective exploration (Fig 3.b). The above approach is explained in detail as follows.

A. Task description:

We apply the proposed framework to two bipedal control tasks: parkour and dive, with varying objectives and extent of multi-modality as shown in table I. Note that the choice of task-vital multi-mode is user-defined, and table I simply reflects our choice for the same. Evident from table I, $J_{\mathcal{T}}$ is task-dependent. Recent attempts in quadruped parkour [2], [3], and locomotion [1], [6] show some well-shaped candidates for $J_{\mathcal{T}}$, albeit case-specific. In general, a reasonable $J_{\mathcal{T}}$ could be hard to design (for instance, the dive task); a compelling unified alternative would be to “track” the oracle’s ϵ -neighbourhood reference to the optimal solution. Hence, we propose minimizing the task-independent surrogate objective: $\tilde{J}_{\mathcal{T}} := \sum_{t=0}^{\infty} \|x_t^\pi - x_t^\pi\|$. $\tilde{J}_{\mathcal{T}}$ ’s applicability is studied in Sec. IV-B and a reward is proposed in Sec III-C for an equivalent maximization objective.

B. Oracle Design

For any locomotion task, a simple heuristic for an oracle would be linearly interpolating the relevant state variables from the initial to the desired goal states. In parkour, to advance along the track, the heading position can be linearly interpolated along the track while adapting to the terrain height, as shown in Fig. 4 (left). For dive, the oracle can linearly interpolate the base height and corresponding rotational DoF from 0° to 360°. Naming this heuristic oracle as Ξ_{li} (Fig. 4), its high ϵ is obvious as the generated ansatz do not consider the system’s inertia and gravity. Hence oracles that capture the dominant dynamics of the hybrid system are required. To this end, we have used a modified version of the simplified Single Rigid Body (SRB) model [25] whose dynamics in the world coordinates are given by:

$$m(\ddot{p} + g) = \sum_{i=1}^2 f_i, \frac{d}{dt}(I\omega) \approx I\dot{\omega} = \sum_{i=1}^2 (r_i \times f_i + \tau_i) = \sum_{i=1}^2 \bar{\tau}_i \quad (6)$$

$$x_{t+1} = Ax_t + But, y_{t+1} = Cx_t, u_t = [f_1, f_2, \bar{\tau}_1, \bar{\tau}_2]^T \quad (7)$$

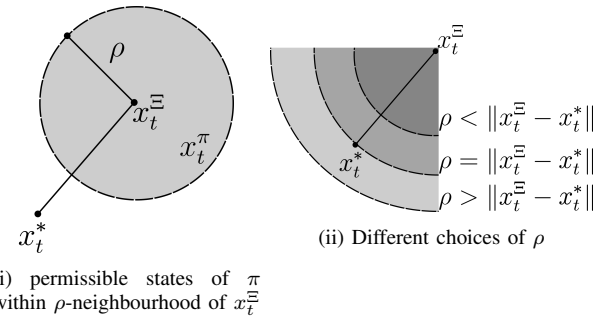


图 2: Oracle 有界探索

π 、 ρ 的允许状态必须满足

$$(x_t^* \text{ is reachable by } x_t^\pi) \implies \rho \geq \|x_t^\pi - x_t^*\| \quad (4)$$

根据定义（式 1.c），由于 $\|x_t^\pi - x_t^*\| < \epsilon$ 是满足式 4 在所有时刻成立的充分选择 ρ ，因此

$$\rho \geq \epsilon \quad (5)$$

ϵ 是 x^π 的最大偏差，低 ϵ 的 Oracle 将生成更接近 x^* 的参考，因此探索可以被限制在紧邻域内，以滤除目标景观中的大多数局部最优。相反，对于高 ϵ 的“差”Oracle，应有足够的搜索空间供 π 围绕 x^π 探索并收敛至 x^* 。根据式 5，当 $\epsilon \rightarrow \infty$ 时，优化是无引导的，因此需要 $\rho \rightarrow \infty$ （标准强化学习设置）。反之，当 $\epsilon \rightarrow 0$ 时，可以选择任意小的 ρ 满足式 5，通过局部化搜索来避免局部最优，同时仍能收敛至 x_t^* 。实践中，由于 ϵ 未知，我们对任意给定的 $(\mathcal{T}, \Xi)$ 进行 ρ 的网格搜索。

2) 任务关键的多模态性：学习解决任务 $\mathcal{T}$ 的策略，可以看作是在任务空间中掌握一组时空轨迹的“束”，对应于 $\psi_{\mathcal{T}} \in \Psi_{\mathcal{T}}$ 。简单任务允许直接构建满足式 1 的 Oracle。然而，复杂且无限时域的任务使得 $\Psi_{\mathcal{T}}$ 难以处理。例如，无限跑酷的 Oracle 需要先验地知道无限长的赛道，这并不实际。为解决此问题，我们将模式定义为保留某些空间和/或时间不变性的有限时空片段。就像在跑酷中，跳跃和腾跃等模式无论位置或时间如何都保持一致。因此，我们定义了一个有限的模式集合 $\mathbb{M}$ ，其时间长度为 Δt ，对 $\mathcal{T}$ 至关重要。每个 $m \in \mathbb{M}$ （如跳跃）可以有连续参数 Ψ_m （如跳跃高度）。那么模式参数集 $\Psi_{\mathbb{M}} := \bigcup_{i=1}^{|\mathbb{M}|} \Psi_{m_i}$ 与任务参数集的关系为 $\Psi_{\mathcal{T}} := \Psi_{\mathbb{M}}^N$ ，其中 N 是时域数量。因此，通过掌握 $\mathbb{M}$ 中的模式（跳跃、腾跃和步速）以及在不同 $\Psi_{\mathbb{M}}$ （速度、距离和高度）上的转换，任务 $\mathcal{T}$ （无限跑酷， $N \rightarrow \infty$ ）可以如图 3.a 所示得到解决。

III. 设计方法

本节介绍了使用 OGMP 进行双足控制任务（跑酷和跳水）的设计方法，如图 3 所示。对于给定任务，我们首先定义任务关键模式——例如跑酷中的跳跃、腾跃和步态——并为每种模式设计参考生成方案。

$\mathcal{T} : J_{\mathcal{T}}$	$(m, \Psi_m) \in (\mathbb{M}, \Psi_{\mathbb{M}})$	
	Mode	Parameters
dive: 360° ip and land	settle ip	$\{\}$ $\{(r, h)\}$
parkour: traverse the track indefinitely	pace jump leap	$\{v\}$ $\{(w, h)\}$ $\{(w, d)\}$

表 I: 任务描述及对应的任务关键多模态。参数可视化见图 4。

（图 3.a）。通过遍历模式参数集，我们利用预言机生成包含多样化行为的自定义数据集，并训练模式编码器以构建用于指令条件化的紧凑潜在空间（图 3.c）。最后，我们训练一个由预言机引导的多模式策略（图 3.d）。在策略优化过程中，定期查询预言机以在线生成参考，将策略的搜索空间限制在参考的局部邻域内，从而实现有效探索（图 3.b）。上述方法详细说明如下。

A. 任务描述：

我们将所提出的框架应用于两个双足控制任务：跑酷和跳水，这两个任务具有不同的目标和多模态程度，如表 I 所示。请注意，任务关键多模态的选择由用户定义，表 I 仅反映了我们的选择。从表 I 可以看出， $J_{\mathcal{T}}$ 是任务相关的。近期在四足跑酷 [2], [3], 和运动 [1], [6] 方面的尝试显示了一些形态良好的候选方案用于 $J_{\mathcal{T}}$ ，尽管这些方案具有特定情境性。通常，合理的 $J_{\mathcal{T}}$ 可能难以设计（例如跳水任务）；一个令人信服的统一替代方案是“跟踪”预言机的最优解 ϵ 邻域参考。因此，我们提出最小化任务无关的替代目标： $\tilde{J}_{\mathcal{T}} := \sum_{t=0}^{\infty} \|x_t^\pi - x_t^\pi\|$ 。 $\tilde{J}_{\mathcal{T}}$ 的适用性将在第 IV-B 节中研究，并在第 III-C 节中提出一个奖励函数以实现等效的最大化目标。

B. 预言机设计

对于任何运动任务，一个简单的预言机启发式方法是从初始状态到目标状态对相关状态变量进行线性插值。在跑酷中，为了沿赛道前进，航向位置可以沿赛道线性插值，同时适应地形高度，如图 4（左）所示。对于俯冲动作，预言机可以对基座高度和相应的旋转自由度从 0° 到 360° 进行线性插值。将此启发式预言机命名为 Ξ_{li} （图 4），其高 ϵ 是显而易见的，因为生成的假设未考虑系统的惯性和重力。因此，需要能够捕捉混合系统主导动力学的预言机。为此，我们使用了简化单刚体（SRB）模型 [25] 的修改版本，其世界坐标系中的动力学方程如下：

$$m(\ddot{p} + g) = \sum_{i=1}^2 f_i, \frac{d}{dt}(I\omega) \approx I\dot{\omega} = \sum_{i=1}^2 (r_i \times f_i + \tau_i) = \sum_{i=1}^2 \bar{\tau}_i \quad (6)$$

$$x_{t+1} = Ax_t + But, y_{t+1} = Cx_t, u_t = [f_1, f_2, \bar{\tau}_1, \bar{\tau}_2]^T \quad (7)$$

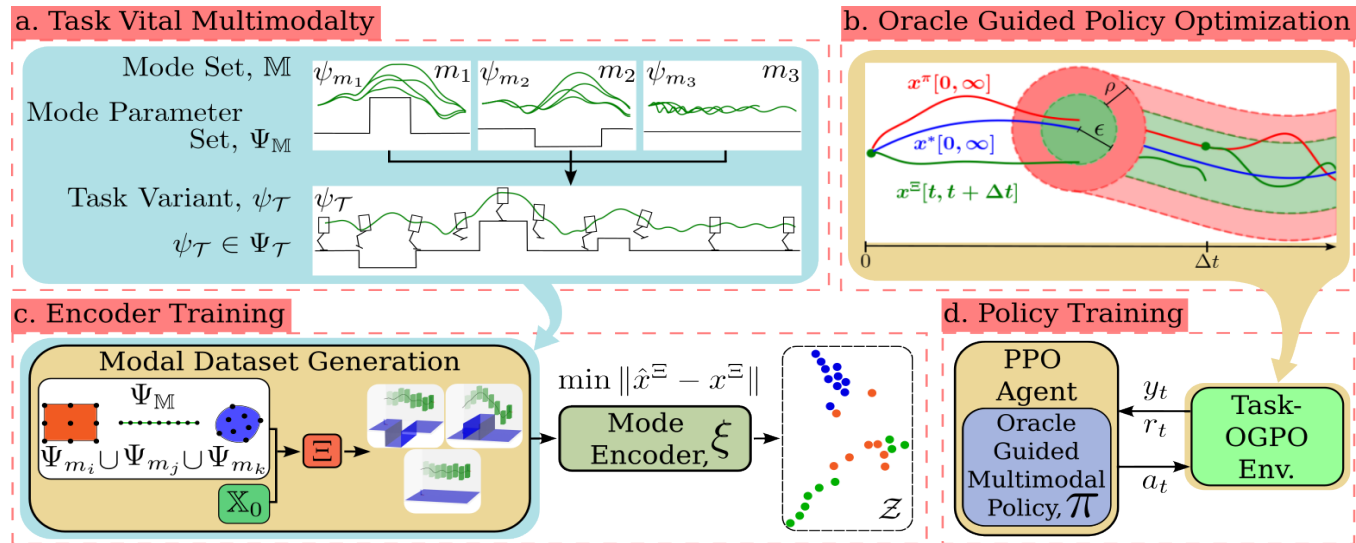


Fig. 3: Overview of the design methodology: a) The breakdown of a task into its mode and mode parameter set b) Guided exploration by bounded permissible state space around the local neighborhood of the oracle's reference c) Mode encoder: an LSTM autoencoder trained on a custom modal dataset by minimizing reconstruction loss d) multi-mode policy trained with oracle guided policy optimization on a task environment

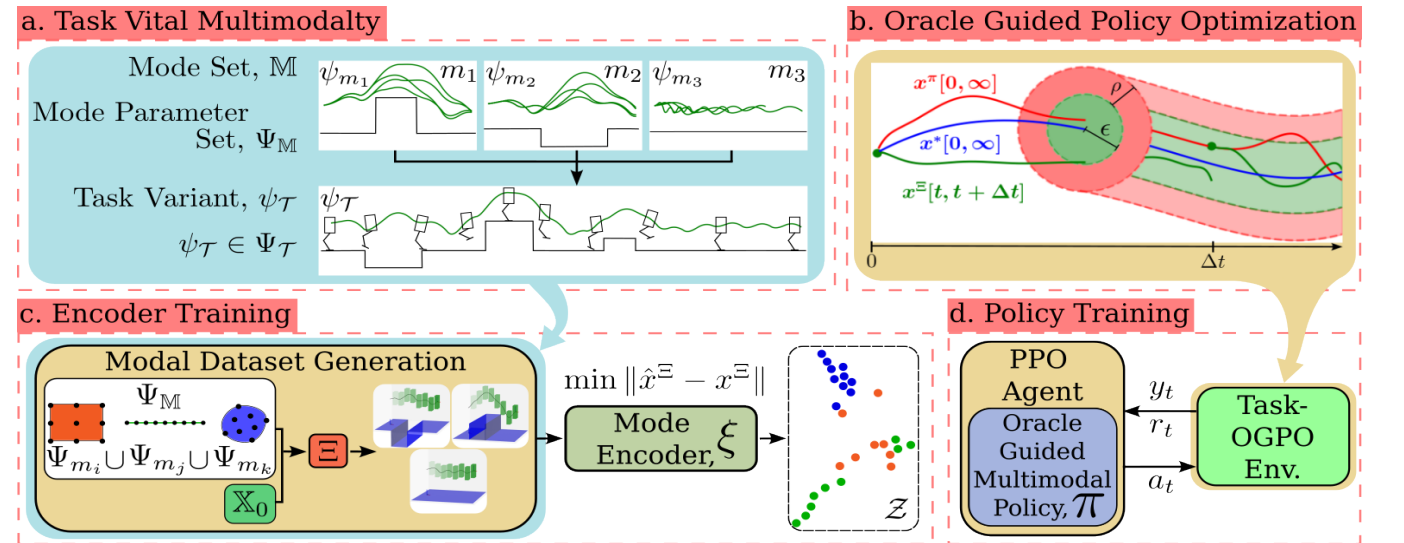


图 3: 设计方法概述: a) 将任务分解为其模式及模式参数集 b) 通过围绕 Oracle 参考局部邻域的有界允许状态空间进行引导探索 c) 模式编码器: 一个在自定义模式数据集上通过最小化重构损失训练的 LSTM 自编码器 d) 在任务环境中使用 Oracle 引导的策略优化训练的多模式策略

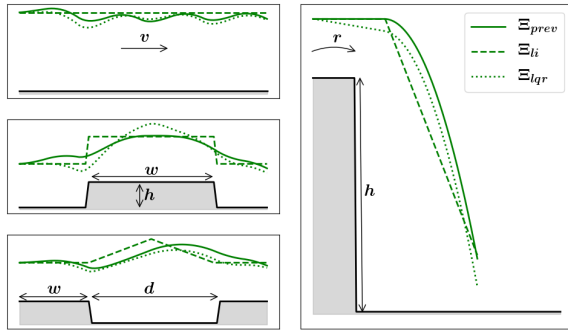


Fig. 4: Base height reference trajectories from various oracles for different modes

where $\ddot{p}$, ω is the robot COM acceleration and the angular velocity, r_i , f_i , τ_i are the position, force and moment vectors from the i^{th} contact point and m , I , g are the mass, moment of inertia and gravity. Typically, r_i is from a predefined contact schedule, leading to time-varying dynamics. Since, by definition, an oracle need not provide realistic control, we define an auxiliary control $\bar{\tau}_i$, encompassing the overall moment, making the dynamics time-invariant. Additionally, the rotation and rotation rate matrices are made constant by considering the average reference orientation over a horizon. Upon these approximations to Eq. 6 and discretization leads to a linear time-invariant (LTI) system over the current horizon, where $x_t \in \mathbb{R}^{13}$, $y_t, u_t \in \mathbb{R}^{12}$ are the gravity-augmented state, relevant output, and control vectors. Thus, oracles can be constructed considering two distinct phases: flight and contact. In flight, $u_t = 0$ as there are no contacts, and during contact, an optimizer of choice can be used to compute the optimal control for a given objective, $u_t = u_t^*$. The reference state trajectory is obtained by applying the corresponding control and forward simulating Eq. 7. Using Ξ_{li} as the reference for a quadratic tracking objective on the

LTI system, optimizing with preview control [29] and LQR results in oracles, Ξ_{prev} and Ξ_{lqr} respectively, having a smaller ϵ than Ξ_{li} as shown in Fig 4

C. Multi-mode Policy:

Mode Encoder: We train a mode encoder, ξ , on diverse locomotion modes to obtain a compact conditioning space ideal for commanding our policy. Similar to [28], the encoder, $z = \xi(x[t, t + \Delta t])$, maps the trajectory space to a latent space ($\dim(z) = 2$). Uniformly sampling from a mode parameter set, Ψ_M and a set of initial states, $\mathbb{X}_0$, we generate a rich and balanced modal dataset by querying the oracle, Ξ as shown in Fig. 3.c. Minimizing the reconstruction loss for a 32 neurons single hidden layer LSTM auto-encoder on the custom dataset generates a set of latent mode points with a structured clustering as visualized in Fig. 3.c.

Mode Conditioned Policy: Our choice of action space for a stationary policy is from [28]. Given the inherent partial observability of the system, for the observation space $o_t \in \mathcal{O}$ we choose $o_t = [\tilde{x}_t, z_t, c_t, h_t]$, where $\tilde{x}_t$ is the robot's proprioceptive feedback, z_t is the latent mode command, c_t is a clock signal [28] and h_t is optional task-based feedback (like terrain scan for parkour). The per-step reward for a task-agnostic surrogate tracking objective is defined as

$$\begin{aligned} r_t &:= r_{\text{track}} + r_{\text{regulation}} \\ r_{\text{regulation}} &:= 0.05e^{-0.01\|u_t\|} - 0.3 \cdot \mathbf{1}(\text{if non-toe contact}) \\ r_{\text{track}} &:= 0.475e^{-5\|er_p\|} + 0.475e^{-5\|er_o\|} \end{aligned} \quad (8)$$

where er_p, er_o are the errors in base position and orientations. Thus, r_{track} minimizes an error in the task space, and $r_{\text{regulation}}$ regularizes the motion for enhancing sim-to-real. Note for the diverse modes across both the tasks (parkour and dive) the reward weights remain the same, hinting a sense of algorithmic robustness that arises from guided learning. The proposed permissible state constraint

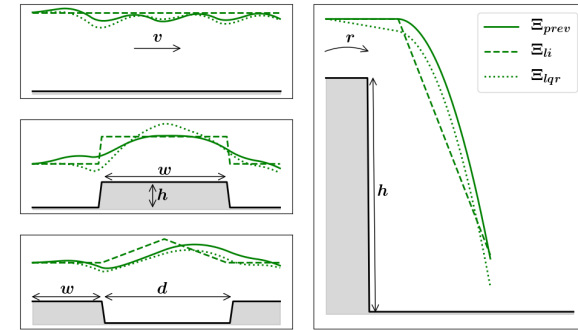


图 4: 不同模式下来自多个 Oracle 的基准高度参考轨迹

其中 $\ddot{p}$, ω 是机器人质心加速度和角速度, r_i , f_i , τ_i 是来自第 i^{th} 个接触点的位置、力和力矩向量, m , I , g 是质量、惯性矩和重力。通常, r_i 来自预定义的接触时序, 导致时变动力学。由于根据定义, Oracle 无需提供现实的控制, 我们定义了一个辅助控制 $\bar{\tau}_i$, 包含整体力矩, 使动力学变为时不变。此外, 通过考虑一个时间窗口内的平均参考方向, 使旋转矩阵和旋转速率矩阵保持恒定。基于这些对公式 6 的近似以及离散化, 在当前时间窗口内得到一个线性时不变系统, 其中 $x_t \in \mathbb{R}^{13}$, $y_t, u_t \in \mathbb{R}^{12}$ 分别是重力增强状态、相关输出和控制向量。因此, Oracle 可以针对两个不同阶段构建: 飞行阶段和接触阶段。在飞行阶段, 由于没有接触, $u_t = 0$; 而在接触阶段, 可以使用任意优化器针对给定目标 $u_t = u_t^*$ 计算最优控制。参考状态轨迹通过应用相应控制并前向仿真公式 7 获得。使用 Ξ_{li} 作为二次跟踪目标的参考,

在 LTI 系统上, 通过预览控制 [29] 和 LQR 优化分别得到 Oracle Ξ_{prev} 和 Ξ_{lqr} , 其 ϵ 小于 Ξ_{li} , 如图 4 所示

C. 多模式策略:

模式编码器: 我们训练了一个模式编码器 ξ , 基于多样化的运动模式, 以获得一个紧凑的条件空间, 便于指挥我们的策略。与 [28], 编码器类似, $z = \xi(x[t, t + \Delta t])$ 将轨迹空间映射到潜在空间 (维度 $z = 2$)。通过从模式参数集 Ψ_M 和一组初始状态 $\mathbb{X}_0$ 中均匀采样, 我们通过查询预言机生成了一个丰富且平衡的模式数据集, Ξ 如图 3.c 所示。在自定义数据集上最小化一个 32 神经元单隐藏层 LSTM 自编码器的重构损失, 生成了一组具有结构化聚类的潜在模式点, 如图 3.c 所示。

模式条件策略: 我们为固定策略选择的动作空间来自 [28]。鉴于系统固有的部分可观测性, 对于观测空间 $o_t \in \mathcal{O}$, 我们选择 $o_t = [\tilde{x}_t, z_t, c_t, h_t]$, 其中 $\tilde{x}_t$ 是机器人的本体感觉反馈, z_t 是潜在模式指令, c_t 是时钟信号, [28] 而 h_t 是可选的基于任务的反馈 (如跑酷中的地形扫描)。用于任务无关的替代跟踪目标的每步奖励定义为

$$\begin{aligned} r_t &:= r_{\text{track}} + r_{\text{regulation}} \\ r_{\text{regulation}} &:= 0.05e^{-0.01\|u_t\|} - 0.3 \cdot \mathbf{1}(\text{if non-toe contact}) \\ r_{\text{track}} &:= 0.475e^{-5\|er_p\|} + 0.475e^{-5\|er_o\|} \end{aligned} \quad (8)$$

其中 er_p , er_o 是基座位置和方向的误差。因此, r_{track} 最小化任务空间中的误差, 而 $r_{\text{regulation}}$ 对运动进行正则化以增强从仿真到现实的迁移。请注意, 对于跨任务 (跑酷和潜水) 的多样化模式, 奖励权重保持不变, 这暗示了引导学习带来的算法鲁棒性。所提出的允许状态约束

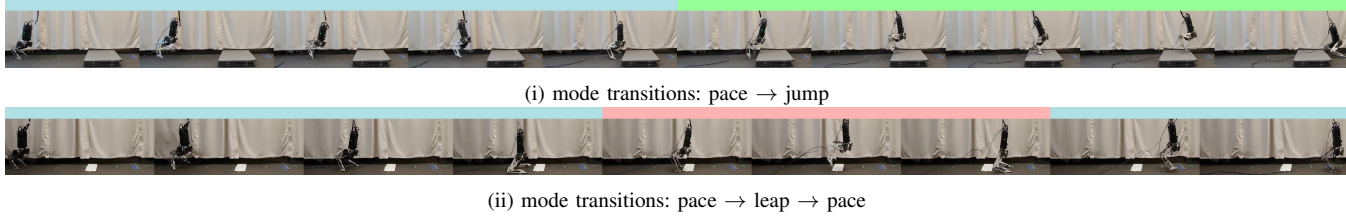


Fig. 5: Keyframes of an OGMP (π_{parkour}) demonstrating mode transitions (left to the right) with the color of the active mode marked: blue for pace, red for leap, and green for jump.

(Sec. II) is programmed as a termination condition, terminate episode:= $\mathbb{1}(\|x_t^\pi - x_t^\Xi\|_W > \rho)$. W_{11} and W_{33} are set to 1 for the parkour and dive tasks, respectively, with the remaining entries as zero. For solving the resulting random horizon POMDP, we use off-the-shelf PPO to train a policy: 128 nodes per layer, 2 layer LSTM network, where each episode is an arbitrary variant of the task $\psi_{\mathcal{T}}$ uniformly sampled from $\Psi_{\mathcal{T}}$.

IV. RESULTS

A. Performance

To achieve extreme agility and mode versatility, a single multi-mode policy is trained per task: π_{parkour} for parkour and π_{dive} for diving. The supplementary video and Fig 1 show π_{parkour} successfully navigating challenging parkour tracks with blocks and gaps placed randomly, demonstrating versatile agility over leap lengths and jump heights. π_{dive} performs omnidirectional flips from different heights and transitions smoothly to landing. Despite lacking a reference for the actuated DoFs, π_{dive} learns an emergent behavior to curl and extend its legs for flips and landings, modulating the torso angular velocity and landing impact. As seen in the video, π_{parkour} significantly deviates from the oracle’s reference to find the optimal behavior, yet results in regularized motion due to the oracle bound. Sim-to-real transfer of π_{parkour} ’s modes and transitions can be seen in Fig 1,5 and supplementary video.

1) *Agility*: For quantitative benchmarking of agility, we report the sample mean of performance metrics: Maximum Heading Acceleration (M.H.A), Froude number (M.F), Maximum Heading Speed (M.H.S), Episode Length (E.L), measured over 100 episode rollouts in table II. We define a test environment with a track length of 10m, an episode length of 400 steps for the parkour, and an episode length of 150 for the dive. In each case, the episode is terminated only if the episode length is reached or the robot falls down (terrain-relative base height < 0.3 m).

metric, units	z_t	z_t, c_t	h_t	h_t, c_t	h_t, z_t	h_t, z_t, c_t
M.H.A	4.7g	3.5g	3.2g	3.6g	3.5g	3.1g
M.H.S (v)	1.4	1.57	1.66	1.74	1.74	1.77
M.F ($\frac{v^2}{g \cdot l}$)	0.48	0.56	0.64	0.69	0.70	0.72
% E.L	0.18	0.43	0.63	0.80	0.66	0.84

TABLE II: Estimated metrics for variants of π_{parkour} with different observation spaces

On average, we find π_{parkour} to reach accelerations of $4.7g$ with heading speeds of 1.77 m/s, and Froude numbers

between $0.48 - 0.72$ while completing 84% of the track as shown in table II. π_{parkour} dynamically advances along the track, avoiding conservative motions with precise foot placements for landing and take-off, leading to agile manoeuvres. The measured Fraude numbers and resulting motion are consistent with [30], where a switch from energy-efficient walking to agile jumping gaits was observed in bipedalism for a value around 0.5.

2) *Mode Versatility*: Since the defined $\Psi_{\mathbb{M}}$ of our training is a compact set, we leverage it to visualize the generalization of π_{parkour} over mode parameters. Dilating $\Psi_{\mathbb{M}}$ and defining higher test ranges for each parameter, we test for both in-domain (ID) and out-of-domain (OD) generalization. Discretizing this test set, we evaluate π_{parkour} and plot the undiscounted returns obtained for blocks and gaps with varying dimensions in Fig 6ii. Different blocks and gaps require jumps and leaps of varying magnitudes, showcasing our policies’ versatility. The training sets Ψ_{jump} and Ψ_{leap} are the regions within the boundary marked in black in each plot. Thus, π_{parkour} shows consistent performance for variants within the black boundary (ID) while also extrapolating its skills by jumping and leaping over unseen terrain variants outside the black boundary (OD) as seen in the supplementary video and Fig 6ii.

B. Ablations and Analysis

Finally, we analyze our choice of surrogate $\tilde{J}_{\mathcal{T}}$ and design choices that impact performance (measured via undiscounted returns). For parkour, a potential true objective, $J_{\mathcal{T}}$, is the displacement along the track. Thus the validity of using $\tilde{J}_{\mathcal{T}}$ can be quantified through its disparity with $J_{\mathcal{T}}$.

Choice of observation space (o_t): Ablation of observation space components (excluding $\tilde{x}_t$) shows that variants with c_t consistently outperform their counterparts (Fig. 6i.a). Variants without h_t are myopic and aggressive, with higher accelerations (table II) as they purely rely on compressed z_t for terrain feedback, making them sub-optimal compared to terrain-aware variants. From Fig. 6i.a, we observe that latent conditioning does not improve performance (see $[h_t, c_t]$ and $[z_t, h_t, c_t]$), hence is purely for analysis and reusability. However, a conditioned π_{parkour} can use the oracle as a closed-loop reactive planner during inference, driving the system to the commanded mode. From Fig.6i.a and b, a similar trend of $J_{\mathcal{T}}$ and $\tilde{J}_{\mathcal{T}}$ shows no disparity caused by observation space variations.

Choice of oracle (Ξ): The three oracles, Ξ_{li} , Ξ_{lqr} , and Ξ_{prev} , have a non-increasing trend in maximum deviation

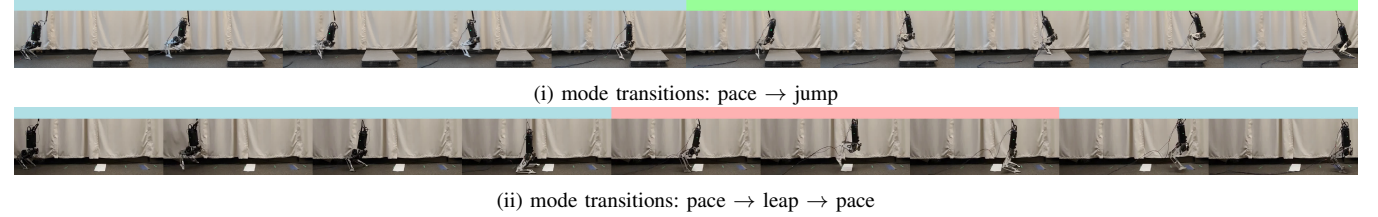


图 5: OGMP (π_{parkour}) 的关键帧, 展示了模式转换 (从左到右), 活动模式的颜色已标记: 蓝色代表步态, 红色代表跳跃, 绿色代表弹跳。

(第二节) 被编程为终止条件, 终止回合: = 1 ($\|x_t^\pi - x_t^\Xi\|_W > \rho$)。 W_{11} 和 W_{33} 分别设置为 1 用于跑酷和潜水任务, 其余条目为零。为了解决由此产生的随机视界 POMDP, 我们使用现成的 PPO 来训练策略: 每层 128 个节点, 2 层 LSTM 网络, 其中每个回合是从 $\Psi_{\mathcal{T}}$ 中均匀采样的任务 $\psi_{\mathcal{T}}$ 的任意变体。

IV. 结果

A. 性能

为了实现极致的敏捷性和模式多样性, 每个任务训练一个单一的多模式策略: π_{parkour} 用于跑酷, π_{dive} 用于潜水。补充视频和图 1 展示了 π_{parkour} 成功导航具有随机放置的障碍物和间隙的挑战性跑酷赛道, 展示了在跳跃长度和弹跳高度上的多功能敏捷性。 π_{dive} 从不同高度执行全方位跳跃, 并平滑过渡到着陆。尽管缺乏对驱动自由度的参考, π_{dive} 学习了一种涌现行为, 即蜷缩和伸展腿部以进行跳跃和着陆, 调节躯干角速度和着陆冲击。如视频所示, π_{parkour} 显著偏离了神谕的参考以找到最优行为, 但由于神谕的约束, 最终产生了规则化的运动。 π_{parkour} 的模式和转换的仿真到现实迁移可以在图 1、5 和补充视频中看到。

1) 敏捷性: 为了对敏捷性进行定量基准测试, 我们报告了性能指标的样本均值: 最大航向加速度 (M.H.A)、弗劳德数 (M.F)、最大航向速度 (M.H.S)、回合长度 (E.L), 这些指标在表 II 中通过 100 个回合的 rollout 测量得出。我们定义了一个测试环境, 跑道长度为 10 米, 跑酷任务的回合长度为 400 步, 潜水任务的回合长度为 150 步。在每种情况下, 只有当达到回合长度或机器人摔倒 (地形相对基座高度 < 0.3 米) 时, 回合才会终止。

metric, units	z_t	z_t, c_t	h_t	h_t, c_t	h_t, z_t	h_t, z_t, c_t
M.H.A	4.7g	3.5g	3.2g	3.6g	3.5g	3.1g
M.H.S (v)	1.4	1.57	1.66	1.74	1.74	1.77
M.F ($\frac{v^2}{g \cdot l}$)	0.48	0.56	0.64	0.69	0.70	0.72
% E.L	0.18	0.43	0.63	0.80	0.66	0.84

表 II: 不同观测空间下 π_{parkour} 变体的估计指标

平均而言, 我们发现 π_{parkour} 能够达到 $4.7g$ 的加速度, 航向速度为 1.77 米 / 秒, 弗劳德数

在 $0.48 - 0.72$ 之间, 同时完成如表 II 所示的 84% 的赛道。 π_{parkour} 沿赛道动态前进, 通过精确的足部落点和起跳避免保守运动, 从而实现敏捷机动。测得的弗劳德数及由此产生的运动与 [30], 一致, 其中在双足运动中观察到从节能步行到敏捷跳跃步态的转变, 其值约为 0.5。

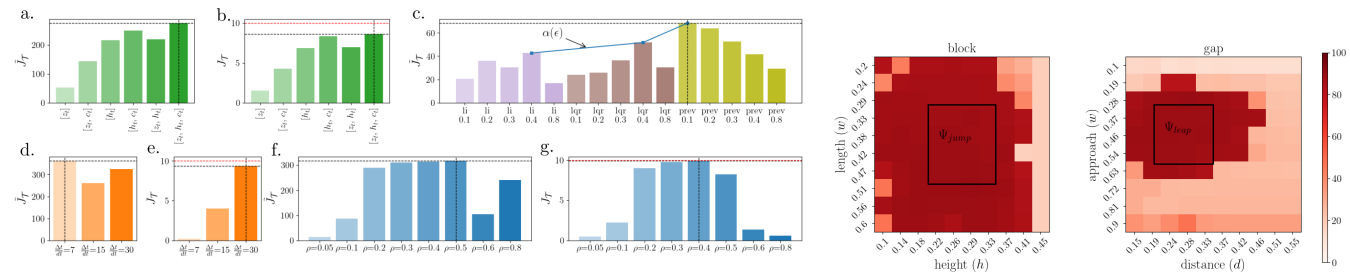
2) 模式多功能性: 由于我们训练中定义的 $\Psi_{\mathbb{M}}$ 是一个紧凑集合, 我们利用它来可视化 π_{parkour} 在模式参数上的泛化能力。通过扩展 $\Psi_{\mathbb{M}}$ 并为每个参数定义更高的测试范围, 我们测试了域内 (ID) 和域外 (OD) 的泛化能力。对该测试集进行离散化处理, 我们评估了 π_{parkour} , 并在图 6ii 中绘制了不同尺寸障碍物和间隙的未折扣回报。不同的障碍物和间隙需要不同幅度的跳跃和腾跃, 展示了我们策略的多功能性。训练集 Ψ_{jump} 和 Ψ_{leap} 是每个图中黑色边界内的区域。因此, π_{parkour} 在黑色边界内 (ID) 的变体上表现出一致的性能, 同时通过跳跃和腾跃跨越黑色边界外 (OD) 未见过的地形变体来外推其技能, 如补充视频和图 6ii 所示。

B. 消融实验与分析

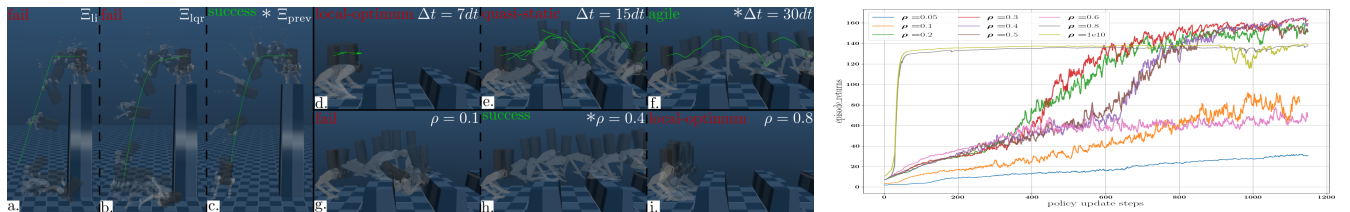
最后, 我们分析了替代 $\tilde{J}_{\mathcal{T}}$ 的选择以及影响性能 (通过未折扣回报衡量) 的设计选择。对于跑酷而言, 一个潜在的真实目标 $J_{\mathcal{T}}$ 是沿赛道的位移。因此, 使用 $\tilde{J}_{\mathcal{T}}$ 的有效性可以通过其与 $J_{\mathcal{T}}$ 的差异来量化。

观测空间的选择 (o_t): 对观测空间组件的消融 (不包括 $\tilde{x}_t$) 表明, 具有 c_t 的变体始终优于其对应版本 (图 6i.a)。缺少 h_t 的变体目光短浅且行为激进, 加速度更高 (表 II), 因为它们纯粹依赖压缩后的 z_t 进行地形反馈, 这使得它们不如地形感知变体。从图 6i.a 中, 我们观察到潜在条件化并未提升性能 (见 $[h_t, c_t]$ 和 $[z_t, h_t, c_t]$), 因此它纯粹用于分析和可重用性。然而, 条件化的 π_{parkour} 可以在推理过程中将 oracle 用作闭环反应式规划器, 从而将系统驱动到指令模式。从图 6i.a 和 b 中, $J_{\mathcal{T}}$ 和 $\tilde{J}_{\mathcal{T}}$ 的相似趋势表明观测空间变化并未造成差异。

oracle 的选择 (Ξ): 三种 oracle, 即 Ξ_{li} 、 Ξ_{lqr} 和 Ξ_{prev} , 在最大偏差上呈现非递增趋势



(i) The estimated surrogate ($\tilde{J}_{\mathcal{T}}$) and true ($J_{\mathcal{T}}$) objectives vs. choice of 1) observation space, (ii) Heatmap of π_{parkour} 's undiscounted returns performing o_t (a and b), 2) oracles with varied permissible state bound, ρ_i (c), 3) horizon length, Δt (d jumps and leaps of varying magnitudes both in and out of and e) and 4) permissible state bound, ρ (f and g). In b, e, and g the upper bound of $J_{\mathcal{T}}$ is $\Psi_{\mathcal{M}}$ while traversing blocks and gaps of different dimensions marked in red.



(iii) Motion traces comparing the choice of 1) oracles (a, b, c) where Ξ_{li} , Ξ_{lqr} fail while only Ξ_{prev} successful in a side flip dive from a 2m block 2) horizon length (d, e, f) where the hyperparameters. longest horizon ($\Delta t = 30dt$) results in the desired agile performance 3) permissible state bound (g, h, i) where the tighter ($\rho = 0.1$) and lenient ($\rho = 0.8$) bounds fail

Fig. 6: Ablations and Analysis

bound, ϵ . As the optimal exploration bound (ρ^*) depends on ϵ , we vary ρ from 0.1 to 0.8 for each oracle. For parkour, different oracles show no significant performance difference (see video results). However, for dive, Ξ_{prev} variants perform significantly better (Fig. 6i.c and 6iii.c). A tighter bound ($\rho = 0.1$) works best for Ξ_{prev} , as it has the lowest ϵ . Conversely, higher exploratory deviation ($\rho = 0.4$) performs best for Ξ_{lqr} and Ξ_{li} . Thus, $\epsilon_{\text{li}} \geq \epsilon_{\text{lqr}} \geq \epsilon_{\text{prev}} \implies \rho_{\text{li}}^* \geq \rho_{\text{lqr}}^* > \rho_{\text{prev}}^*$, affirming Eq 5.

Choice of oracle's horizon (Δt): We evaluate policy performance across different horizons: $\Delta t = 7dt$, $15dt$, and $30dt$ for parkour. The shortest horizon, $\Delta t = 7dt$, leads to a myopic behavior, maintaining a high $\tilde{J}_{\mathcal{T}}$ but low $J_{\mathcal{T}}$ as it remains stationary by exploiting the high replanning frequency (Fig. 6iii.d and 6i.d, e) as also observed by [13]. Although advancing forward, $\Delta t = 15dt$ fails to anticipate farther terrain, resulting in quasi-static maneuvers (Fig. 6iii.e). Conversely, $\Delta t = 30dt$ enables the robot to leap efficiently from block to block, demonstrating agility and achieving the optimal outcome (Fig. 6iii.f). Increasing Δt aligns $\tilde{J}_{\mathcal{T}}$ with the true task objective, $J_{\mathcal{T}}$ (Fig. 6i.d, e).

Choice of permissible state bound (ρ): For Ξ_{prev} in parkour, we varied ρ from 0.05 to 0.8 (Fig. 6i.f and g). We found an optimal $\rho^* = 0.5$ with performance decreasing away from this value. For $\rho < \rho^*$, the optimal solution may lie outside the ρ -neighborhood of x^{Ξ} (Fig. 6iii.g and Fig. 6iv). Conversely, $\rho > \rho^*$ increases local optima within $\epsilon + \rho$, leading to sub-optimal solutions (high $\tilde{J}_{\mathcal{T}}$), seen in Fig. 6iii.i for $\rho = 0.8$, where the policy stagnates without advancing (low $J_{\mathcal{T}}$). Training curves in Fig. 6iv show that $0.1 \leq \rho \leq 0.6$ converges to the global optima, while the rest settle at local optima. Note that $\rho \rightarrow \infty$ is standard PPO

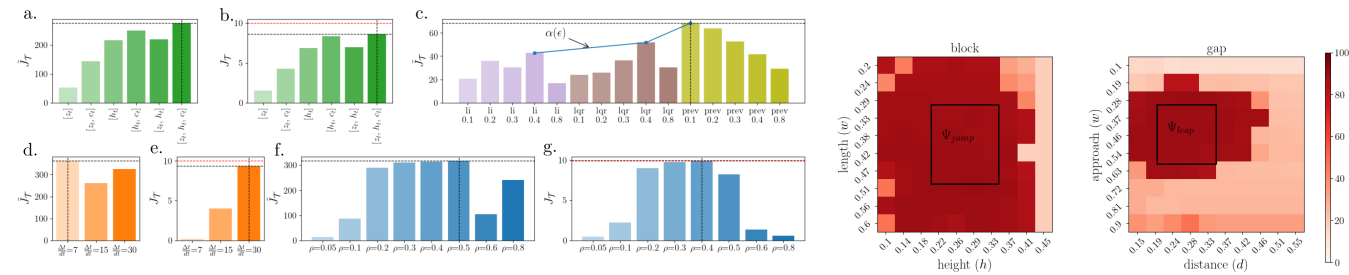
as it optimizes $\tilde{J}_{\mathcal{T}}$ unguided by Ξ . Vanilla unguided PPO ($\rho = 10^{10}$) falls into the same local optima as $\rho = 0.8$. Thus, oracle-guided optimization improves standard PPO by escaping local optima with the right choice of ρ .

V. CONCLUSION

This paper introduces a framework for guided policy optimization through prior-bounded permissible states and task-vital multi-modality to tackle complex tasks. A single OGMP (per task) successfully solved agile bipedal parkour and diving, showcasing versatile agility. Future work will aim to extend to contact-rich open-world loco-manipulation tasks. Furthermore, by restricting the reachable states to a subset of the state space, we forgo the possibility of OGMP being a global policy. Consequently, any state outside the ρ -neighbourhood of the oracle's references may result in failure. Since current Deep RL methods lack any global convergence guarantees, this is nonrestrictive but highlights the need for future extensions for stronger algorithms.

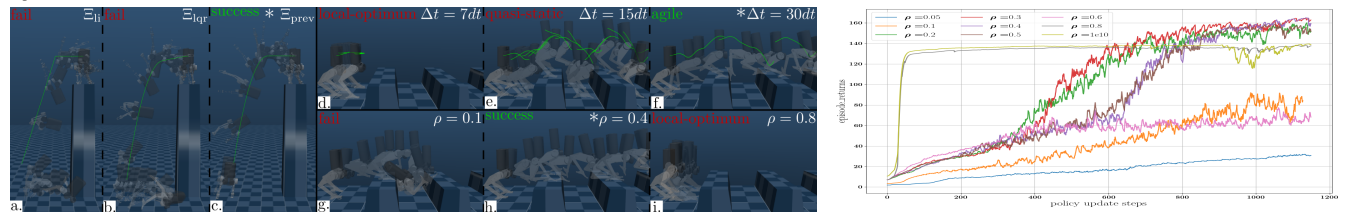
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(i) 估计的替代目标 ($\tilde{J}_{\mathcal{T}}$) 与真实目标 ($J_{\mathcal{T}}$) 随以下因素的变化: 1) 观测空间的选择, o_t (a 和 b), 2) 具有不同允许状态边界的预言机, Ξ_{ρ} . (c), 3) 规划时间长度, Δt (d 和 e), 以及 4) 允许状态边界, ρ (f 和 g)。在 b、e 和 g 中, $J_{\mathcal{T}}$ 的上界用红色标出。(ii)

π_{parkour} 在穿越不同尺寸的障碍块和间隙时, 执行不同幅度的跳跃和腾空动作 (包括在 $\Psi_{\mathcal{M}}$ 内和外的) 的未折扣回报热力图



(iii) 运动轨迹对比: 1) 预言机的选择 (a, b, c), 其中 Ξ_{li} 、 Ξ_{lqr} 失败, 仅 Ξ_{prev} 成功完成从 2 米高障碍块侧身俯冲; 2) 规划时间长度 (d, e, f), 最长规划时间 ($\Delta t = 30dt$) 实现了期望的敏捷性能; 3) 允许状态边界 (g, h, i), 较紧 ($\rho = 0.1$) 和较宽松 ($\rho = 0.8$) 的边界均失败。(iv) 在固定 PPO 超参数集下, 不同 ρ 的训练曲线

图 6: 消融实验与分析

边界, ϵ 。由于最优探索边界 (ρ^*) 取决于 ϵ , 我们对每个预言机将 ρ 从 0.1 变化到 0.8。对于跑酷任务, 不同预言机未表现出显著性能差异 (见视频结果)。然而, 对于俯冲任务, Ξ_{prev} 变体表现显著更优 (图 6i.c 和 6iii.c)。较紧的边界 ($\rho = 0.1$) 对 Ξ_{prev} 效果最佳, 因其具有最低的 ϵ 。相反, 较高的探索偏差 ($\rho = 0.4$) 对 Ξ_{lqr} 和 Ξ_{li} 表现最佳。因此, $\epsilon_{\text{li}} \geq \epsilon_{\text{lqr}} \geq \epsilon_{\text{prev}} \implies \rho_{\text{li}}^* \geq \rho_{\text{lqr}}^* > \rho_{\text{prev}}^*$, 确认了公式 5。

预言家视野范围的选择 (Δt): 我们评估了不同视野范围下的策略性能: $\Delta t = 7dt$ 、 $15dt$ 和 $30dt$ 用于跑酷任务。最短的视野 $\Delta t = 7dt$ 导致短视行为, 保持高 $\tilde{J}_{\mathcal{T}}$ 但低 $J_{\mathcal{T}}$, 因为它通过利用高重规划频率保持静止 (图 6iii.d 和 6i.d,e), 这也被 [13] 观察到。虽然向前推进, $\Delta t = 15dt$ 未能预判更远的地形, 导致准静态操作 (图 6iii.e)。相反, $\Delta t = 30dt$ 使机器人能够高效地从一块砖跳到另一块砖, 展现出敏捷性并达到最优结果 (图 6iii.f)。增加 Δt 使 $\tilde{J}_{\mathcal{T}}$ 与真实任务目标 $J_{\mathcal{T}}$ 对齐 (图 6i.d,e)。

允许状态边界的选择 (ρ): 对于跑酷中的 Ξ_{prev} , 我们将 ρ 从 0.05 变化到 0.8 (图 6i.f 和 g)。我们发现了一个最优的 $\rho^* = 0.5$, 性能在此值附近下降。对于 $\rho < \rho^*$, 最优解可能位于 ρ 邻域之外 (图 6iii.g 和图 6iv)。相反, $\rho > \rho^*$ 增加了 $\epsilon + \rho$ 内的局部最优, 导致次优解 (高 $\tilde{J}_{\mathcal{T}}$), 如图 6iii.i 中 $\rho = 0.8$ 所示, 策略停滞不前而不前进 (低 $J_{\mathcal{T}}$)。图 6iv 中的训练曲线显示, $0.1 \leq \rho \leq 0.6$ 收敛到全局最优, 而其余则停留在局部最优。注意 $\rho \rightarrow \infty$ 是标准 PPO。

因为它优化了 $\tilde{J}_{\mathcal{T}}$ 而不受 Ξ 引导。无引导的普通 PPO ($\rho = 10^{10}$) 陷入了与 $\rho = 0.8$ 相同的局部最优。因此, 预言家引导的优化通过正确选择 ρ , 帮助标准 PPO 逃离局部最优。

V. 结论

本文介绍了一种通过先验有界允许状态和任务关键多模态性来引导策略优化的框架, 以应对复杂任务。单个 OGMP (每项任务) 成功解决了敏捷双足跑酷和跳水问题, 展现了多功能的敏捷性。未来工作将致力于扩展到接触丰富的开放世界移动操作任务。此外, 通过将可达状态限制在状态空间的子集内, 我们放弃了 OGMP 成为全局策略的可能性。因此, 任何处于 oracle 参考的 ρ 邻域之外的状态都可能导致失败。由于当前的深度强化学习方法缺乏全局收敛保证, 这一限制并非严格约束, 但突显了未来需要更强算法扩展的必要性。

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